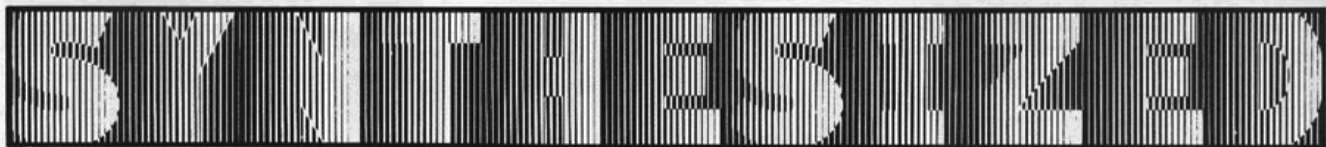




PHOTOGRAPHS BY ARLENE KORETSKY



VIDEO

What Robert Moog and Donald Buchla did for audio in the sixties, Steve Rutt did for video in the seventies. Rutt designed the first coherent, relatively low cost synthesizer. He created a kind of electronic animation capable of modifying an image from a camera and of generating an abstract image without a camera. The Rutt-Etra [RE] synthesizer is in many ways parallel to an audio synthesizer, and it is not difficult for a person who knows sound synthesis to familiarize himself with this new technology. Both devices are real time, voltage controlled systems, and they share many of the same control voltage generators.

by Randy Cohen

The Rutt-Etra developed out of a collaboration between an engineer, Steve Rutt, and an artist, Bill Etra, much as the original Buchla system was influenced by the interaction between Don Buchla and composer Morton Subotnick. Etra's studies had ranged from bio-physics, through the various stage crafts, to still photography, to film documentary work. The film work lead him to docu-

mentary work using portable video gear. His breakthrough came when he turned the video camera to face its own monitor and discovered the joys of video feed-back. This excited Etra about the possibilities of video art. Steve Rutt grew up as a classic American basement kid, building amplifiers and mixers



PHOTOGRAPH BY ARLENE KORETSKY

Steve Rutt

before he entered the sixth grade. His interest in audio engineering reached its zenith at the Yip-In in Central Park in 1968. Rutt did all of the sound work for this cultural high-water, running all of his gear from a 3,000 watt generator in his car.

When these two began their partnership, Rutt was occupied designing industrial electronics, building control gear to monitor the passage of coal down the chutes, dummy tv

cameras for department store security, and gigantic strobe lights used to check details in the manufacturing of steel. Etra was working in the experimental tv lab of WNET in New York, spending much of his time using the Paik-Abbe synthesizer, and becoming increasingly frustrated by its limitations. Etra received some funding from David Loxton, the director of the tv lab, to collaborate with Rutt on the design and construction of a device that would overcome many of the limitations of the Paik-Abbe machine.

The system they developed drew on the advances made by other designers. In addition to Nam June Paik and Shue Abbe, they were influenced by the work of Ben Lapasky, the genius of Cherokee, Iowa, who, in the 1940's, was doing synthesis work using oscilloscopes. Etra also mentioned the Ernie Kovacs show, which gave the first public



PHOTOGRAPH BY LOUISE ETRA

Bill Etra

exposure to modified video images.

Since the video signal and the information it carries are far more complex than audio transmission, they were faced with a vast array of problems. Also, Etra emphasizes that the body of video technology was built to record the real world, and not to alter that imagery. He adds that television technology developed out of radio and tended to discourage the techniques of the film world.

As an artist, Etra was particularly pleased that they were able to develop a real time system. This was a radical development, unavailable in film, animation, or computer work, and it placed the artist in a new and attractive position. Less desirable were the economics of the Rutt-Etra system. Although it costs a fraction of any comparable system, it is still expensive technology. Etra estimates the cost of his own studio, built around the RE and including many peripherals, at over \$40,000. This put great limitations on the numbers of people who would have access to this field, and influenced the ultimate resting place for the machines. Rutt has built systems for WNET in New York, EUE Screengems, the Media Studies Center of the State University of New York at Buffalo, the Art Institute of Chicago, Ontario College of Art, and studios in Venezuela and Australia.

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Bill Etra now devotes his time to the art and not the technology of video. His studio includes, in addition to the RE, the Hearn synthesizer, another voltage controlled machine that performs colorizing functions on the RE's monochrome image, as well as switching, mixing, and keying. Etra has added digital equipment, employing it in a manner similar to that used in many electronic music studios. These small computers, an Altair 8000 and a PDP 11, provide control information to the analog equipment. The studio also contains various peripherals, including a programmable Techtronics scope, cameras, monitors, and tape decks. Etra's

ease and sophistication as he moves from synthesizer to computer console, to tape deck, make one forget the elaborate technology he is handling. He plays with these toys in the best sense, obviously enjoying his work as he spins out image after image, bathing the eyes in color.

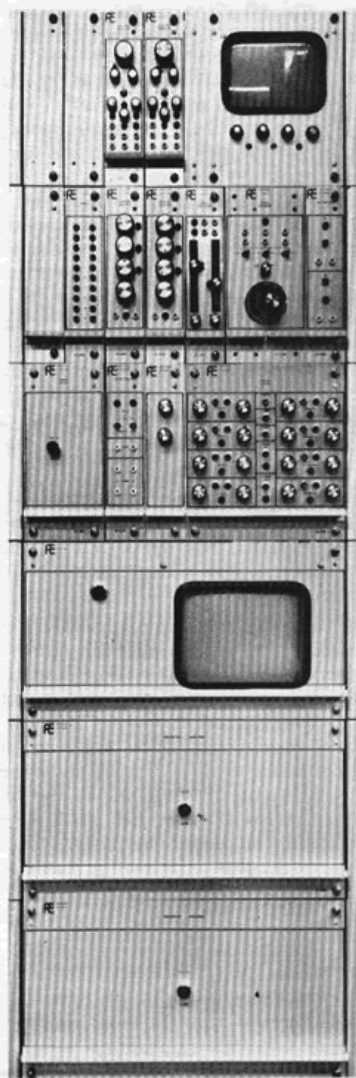
Steve Rutt entered into their partnership with an attitude of "We just build 'em; we don't fly 'em", but his current work, at EUE Screengems, has been creating images, and not designing hardware. Electronic music became prominent in commercials because it provided an economical way to create novel, easily recognized sounds, and Rutt has found a similar situation for his video images. He synthesized fleas and ticks to be dispelled by the Hartz Mountain collar. He warped the image of Dick Martin for a G.E. television spot and sent a stream of words down the tubes for Drano. The titles for the news program in the film *Network* are a Rutt job, as is the new logo for *The Edge Of Night*. When

he was called upon to synthesize the dreaded cough control center for Vicks, the ad agency had three doctors in attendance to assure biological accuracy. He has done in-house work for the Merrill Lynch video network and Atlantic records. He created the abstract imagery for Ophelia's mad scene in the mercifully short lived *Rockaby Hamlet*. These images were transferred from video to film and then projected against a curtain of dry ice smoke, on Broadway. In fact, where ever logos lurch across the screen, chances are that Rutt will be in the background somewhere, usually giggling.

Rutt is presently interested in getting back to design work and is soliciting comments from users of his synthesizer to guide him in designing new video gear. His immediate project is the Image Repositioner, marketed by the Professional Video Group. This device can isolate any portion of a tv image, even if it is already recorded on tape, and move it to any new spot on the screen.

The Rutt-Etra Video Synthesizer

A tv image is built up out of a set of parallel horizontal lines, called the raster, that scan across the screen. The Rutt-Etra can deflect the raster to reconstruct an image. The original image enters the RE on the *Display Unit* and is altered via the functions provided on the *Display Control Unit* (DCU). The DCU offers both manual and voltage controlled manipulation of height, width, depth, intensity, and horizontal and vertical centers. The control voltage functions are housed in the *Animation Module*. The final image is shown on a *High Resolution Display* where it is then shot with a camera and treated as an ordinary video signal. The RE is a monochromatic system; color and other effects are added later. The control voltage generators in this machine can be augmented or replaced by those found in many audio synthesizers. The RE also can be interfaced with digital equipment.



Part Two: Heavy Users

If technically video synthesis parallels electronic music, artistically the parallel breaks down. Listening is very different from looking. Not only is visual information more complex than audio, the problems of form, change over time, transition, and detail are also very different. Visual imagery is embodied with a different set of connotations than is musical imagery. Visual work draws on a narrative tradition that is alien to music. One conceptualizes visual events in a manner quite unlike the experience of listening to music. Thus, visual synthesis does not create a sort of electronic music for the eyes. There can be no visual equivalent to *Hymen*, nor is there any simple musical translation of a tape by Nam June Paik. And the set of visual problems is approached in a unique way by each video artist.

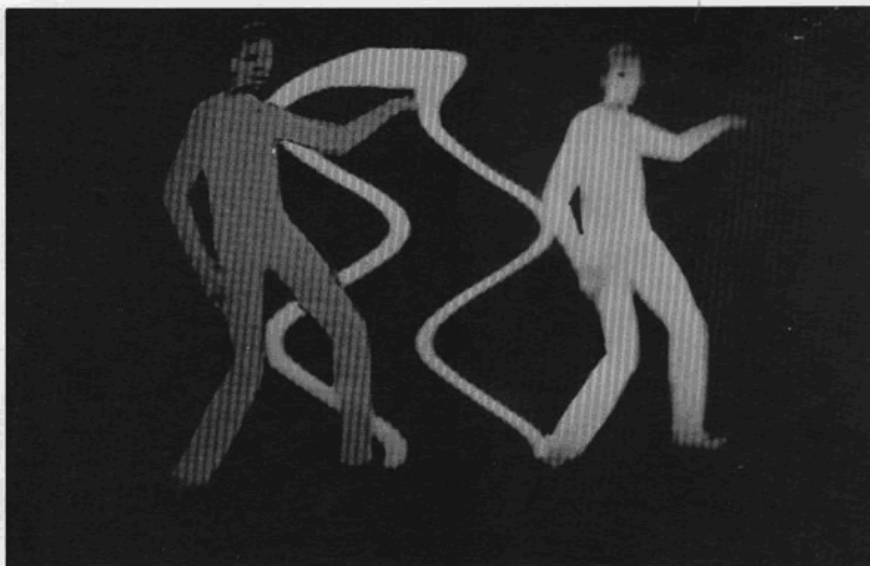
Steve Rutt suggests that much of the best work with his machine treats video synthesis as only a single element of a piece, much as lighting and set designs are only single elements in a theatrical work. The sculptor, Doris Chase, takes a similar view, saying: "I use video as one of my tools, not as *the* tool. When I have a form I want to go further with, I take it to the machine." She creates work in wood, metal and fiber glass, and she sometimes incorporates her sculptures in films, dance works, and video tapes. Regardless of the media used to manifest her ideas, her work carries an integrity and a unity that evidences a powerful artistic vision across the media she employs. It is clearly Ms. Chase in control, not the machines. She does not get seduced by the technology. She describes her approach to video, saying: "I approach the screen as a painter, and work in a plastic manner, treating the edge of the screen as though it were the frame of a moving painting."

Tom DeWitt draws on a background in film, and a familiarity with audio synthesis. He is particularly enthusiastic about the relationship between electronic music and synthesized video:

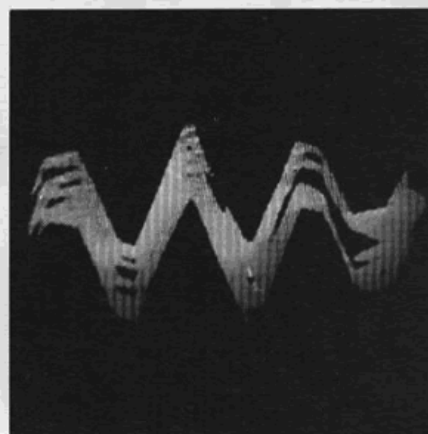
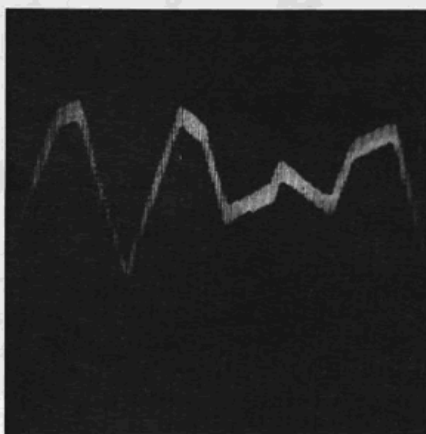
"The Rutt system is an analog voltage controlled device and shares many common characteristics with a Moog, Buchla, etc. It might be held in disdain by digital nuts, but I have always felt that an analog is worth a thousand calculations. Many of the control functions used in audio synthesis can be used to generate pictures. For example, the harmonic structures of music can be designed according to polar co-ordinates in positioning an image on the screen. Incidentally, Rutt's box is still one of the best ways to perform repositioning and is becoming a tool for titling in many studios."

DeWitt uses the RE as a single element in very elaborate works. He displays great technical sophistication and is yet another in the current crop of artist-builders.

Most video artists emphasize the newness of this medium, pointing out that it is very



Above: From Doris Chase's "Dance Nine", dancer: Gus Solomon.

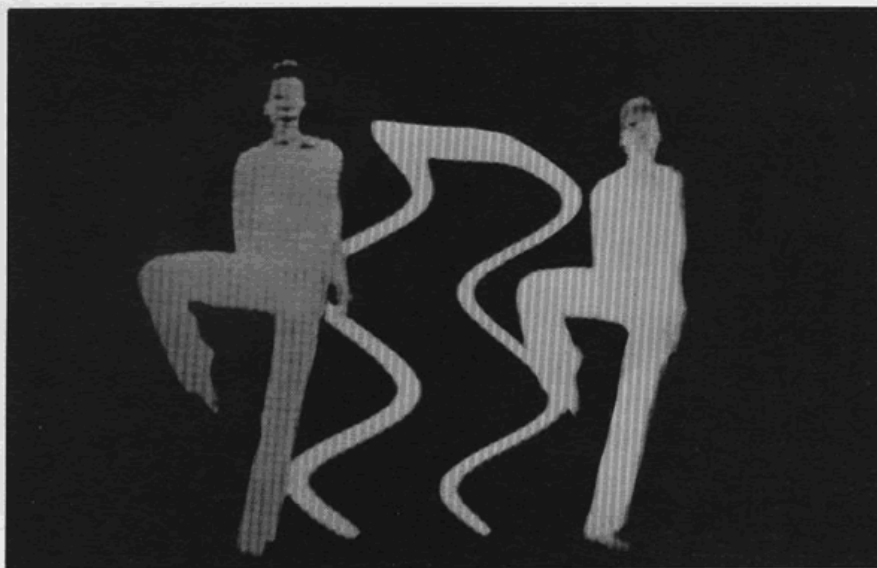


"... Much of the work we do will
... this should be a period of great

much in its formative stages. While music has a tradition stretching back thousands of years, the visual artist has only been able to work with images in motion for, at most, the last fifty years. Bill Etra stresses that the individual artists have had synthesizers for only the last six or seven years, while a contemporary musician may have been working with his instrument for decades. Video artist Laurie Spiegel notes the small number of people in the field and looks forward to the time when the machines will have become sufficiently inexpensive and decentralized to create a body of amateur artists working in their homes. She hopes to see the video synthesizer become as common as the piano in the parlor. However, this argument: "It's only in its primitive stages", must be viewed with a certain skepticism. The composers of ancient India did not view their own work as primitive, and neither did their audience. Mozart did not comment on his

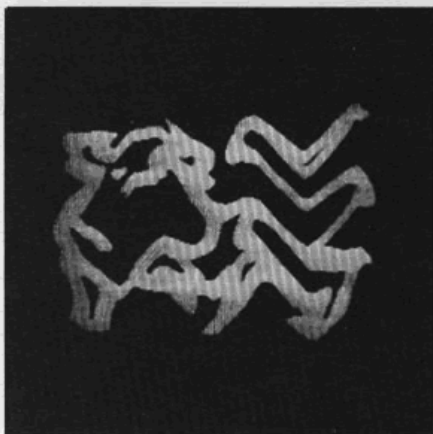
own work: "It ain't bad for primitive work, but once everybody gets at these new pianos, we'll get the really hot stuff, like Beethoven." Ultimately, the work of the present generation of video artists must be evaluated on its own merits, and not simply as the precursors of the next century's really good work.

The most charming thing about video synthesis may be its ability to place flickering light inside of a box in the living room, an effect previously unique to the fireplace. One can gaze into this hearth of the future and let the mind wander. Etra refers to certain of his works as "video wall paper" in the manner of Satie, and Nam June Paik has base a witty and insightful installation on the coincidence between tv sets and aquariums. These ideas suggest that *Charlie's Angels* could be replaced by tropical fish, *Police Woman* by an oven with a glass door, and Johnny Carson by a washer-dryer. The ultimate video work might well be a tape of a crackling fire



PHOTOGRAPHS BY JAMIE NEWMAN

Below: A section from the Merrill Lynch Video Network logo.



PHOTOGRAPHS BY ARLENE KORETSKY

take instruction from musical ideas. . . .
cross fertilization between the arts. . . ."

displayed on a monitor that is placed in a fireplace and set ablaze, and then taped and shown on a second monitor, itself placed in yet another fireplace and set aflame, and then taped. . . .

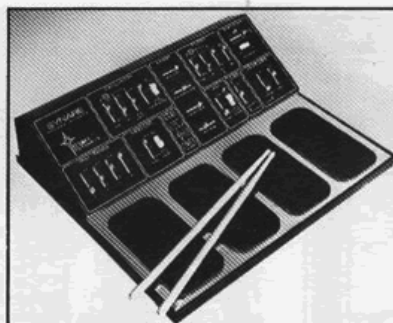
Video synthesis has vast potential to become the ultimate video game, more exciting than Pong, more breath-taking than Death Race, more elaborate than Space War. For the first time, the home audience could re-assert control over their tv sets, striking back at the networks. They could maul Mary Tyler Moore, twist Tom Snyder, or turn Howard Cosell into a quivering mass of idiot globules, just like the networks do.

Since broadcast television has not been overly responsive to the works of a Doris Chase or a Bill Etra, video artists have been facing a distribution problem even more severe than that of the contemporary composer. The new video discs may create a wider audience for these people, but that is in the

future. The current conditions of the field are those of continuing research and experimentation, both technically and artistically. Tom DeWitt praises the contribution of Steve Rutt, whom he refers to as the "most unusual engineer in video-land", and the "continuing innovation of his former partner, Bill Etra." DeWitt assesses the present situation:

"We filmmakers, video artists and computer graphics freaks are just getting into the realm of time change, something that musicians know about from the beginning of their studies. It is an exciting development in visual art to include change. Much of the work we do will take instruction from musical ideas, and it is not surprising to find that initially we are working quite well in music studios. I have lashed my Rutt display up to a Moog, Serge, EMu, Aries, and computers programmed for music with great results. This should be a period of great cross fertilization between the arts." ~~~~

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TANGERINE

DREAM

Tangerine Dream's recent U.S. tour may in many ways be observed as a test.

The question would be, can an electronic music group haul themselves,

their equipment, a laser set-up, and a laserist around the country to halls seating 1500 to 5000 people and return home solvent? The answer is yes. In fact, most dates were sold out. Of course it helps to have many albums out and ads in the national music press announcing the tour schedule, but nevertheless, it's never quite been done before.

by Doug Lynner

Doug Lynner: *I understand that Tangerine Dream originally started from a much more traditional "rock band" format. Would you comment on your beginnings and how you became identified with synthesizers?*

Edgar Froese: Well, the thing was that we felt we couldn't do our music just playing drums and guitars. We had something different in mind and with conventional instruments we felt we couldn't do it because of the limited range of sounds. And so there was coming up the possibility to use synthesizers and instruments that allow you to collect sounds the way you want, not just given sounds from conventional instruments. We started in a very small way with just a little synthesizer first and through the years we were able to try bigger ones and more expensive ones and try different possibilities.

Doug: *Were there any groups or people that influenced your move to synthesizers?*

Edgar: No group.

Chris Franke: No, we actually got the first stuff in Germany because we were in the process of extending our instruments. Conventional instruments are just not enough. Actually, we started using synthesizers, not because they were around and something new, but we thought we had the knowledge and the theory to do much more than just make sound effects. We want to be an explorer in sound.

Peter Baumann: The electronics were just a

result of other things we were doing to make sound. Electronics were a result of that because it was more easy to handle and more universal, even in Germany.

Doug: *Is the introduction of keyboards part of your attempt to be accessible to a popular audience?*

Peter: It doesn't happen by the thought that we want to sell a million albums. Of course we want to sell it, but it comes after the musical development. First there was the musical development of going very far away from what was usual to us. We had to make a very, very big circle to be able to express ourselves in a language that is understandable. That's why we had to make this change.

Doug: *Language is something that I am interested to talk to you about. So far, you've not been doing vocals. Do you plan to use them in the future?*

Edgar: We may use them but definitely not in the normal way.

One thing is that we are from Germany and what we could say in English is not that understandable so why should we sing in bad English?

Peter: Another thing is that singing has a very direct association and the music we are doing leaves very much space for interpretation. People have different access to the music. I think for instance, with Rod Stewart, there are not very many different ways to access. There is only one. Using the voice with special fixed lyrics would limit

that very much. We might do it one day for a special purpose. We would use the voice as another instrument or sound source.

Doug: *What kind of response did you get from American audiences on your trip?*

Peter: Very good.

Doug: *Did they seem open to your music?*

Peter: I think you can't say it that easily. It's very difficult to distinguish. Some people say our electronic music is chance. We realized, especially yesterday, (Santa Monica Civic Concert) that once you have a good relationship to the audience you are able to go really very far out and play things that under normal circumstances they would never accept. But just because you have a good correspondence, they are listening to you. They accept you and they accept what you have to say. They have a feeling for whether you're standing to what you are saying or if you are not. And if you are, they like it.

Doug: *From a live performance viewpoint, what directions do you think that synthesizer designers should work in?*

Edgar: A lot of instruments are very unuseable because the design and the way of handling the instrument is totally senseless. A lot of times we've bought an instrument with little knobs you can't find or switches you can't find.

Chris: There are three aspects which have to be dealt with very fast. Programming; changing programs in performance, which